

AI-POWERED INTELLIGENT EDUCATION SUPPORT SYSTEM

25-26J- 402

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology, specializing in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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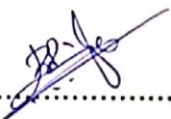
August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


.....

Signature of the supervisor:

Date...29/08/2025

ABSTRACT

One of the strong predictors of student performance at school is student attendance, and yet, the majority of schools of Sri Lanka do not have an intelligent system of monitoring, analyzing, and predicting attendance trends efficiently. This project, Attendance Trend Analyzer, considers solving this with the development of a data-driven system that captures, aggregates, and analyzes student attendance of the student on grade, section, and subject level, and also on school term level. Using state-of-the-art time series forecasting models of ARIMA and LSTM, the system predicts attendance trends of upcoming days as well as identifies anomalies and hence empowers attendance risk identification much prior to time. Moreover, the system performs contextual correlation analysis by superimposing attendance trends with tests occurring in academics, events, holidays, and other external events and presents wiser information beyond attendance data such as attendance. The aggregated data will be represented as the heatmap, graphs, and interactive dashboards such that school administrators as well as teachers are able to make proactive data-driven decisions and intervene long before the student suffers due to low attendance or lack of performance. Expected results are identification of at-risk students, better attendance pattern knowledge, and prescriptive advice on interventions such as ones on one counseling, focused support initiatives, or advisement for the student for their increasing participation and school performance. The project provides an illustration of practical application of use of IT, data analysis, and use of artificial intelligence on its application for education for better monitoring of student participation and performance. The system scales and can install itself on top of line of schools and school terms. Using the power of predictive modeling, anomaly detection, and actionable visualization, the Attendance Trend Analyzer provides an innovation solution which shatters the attendance monitoring paradigm. It not only helps administrators make data-driven decisions but also has the advantage of identifying risk factors for attendance at an early point, delivering customized interventions, and enabling higher student participation and accomplishment.

Keywords: Attendance Analytics, Predictive Modeling, Anomaly Detection, Educational Data, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligent
ML	Machine Learning
ARIMA	Auto-Regressive Integrated Moving Average
LSTM	Long-Short Term Memory
GRU	Gated Recurrent Unit
RNN	Recurrent Natural Network

1.INTRODUCTION

Learning holds the key for social and economic development, and for learners of Sri Lanka, it has a decisive role to play for their futures[1]. Due to the phenomenal rise of technology, the long-standing methods of education struggle more with implications of learners of the digital era. Traditional curricula, as with learning processes, and scarce infrastructures for feedback never are able to cope with variations of speed of learning, motivation, and access. Meanwhile, the COVID-19 pandemic has gone further to emphasize the need for flexible and dynamic solutions for learning, both highlighting and emphasizing traditional ones.[2], [3]

In order to overcome these challenges, this project suggests the development of an AI-Powered Intelligent Education Support System (IES). The system possesses several intelligent modules such as an Adaptive Learning Path Generator, an Academic Risk Prediction System, a Student Stress Level Prediction Module, and an Attendance Trend Analyzer. By implementing the use of artificial intelligence (AI) and machine learning (ML) technologies, the system will enhance learning attainment, assist teachers with decisions, and intervene on target on risk-learners at the right time.[4] The innovation of the project includes designing the entire learning ecosystem which not only fosters academic achievement, inclusiveness, as well as learner emotional well-being and motivation.[5]

In the Sri Lankan situation, the system plays its central role. The education system of the country is extremely exam-oriented with immense pressure on the child for performance at the country-level tests like the Grade 5 Scholarship Examination, the G.C.E. Ordinary Level (O/L), and the G.C.E. Advanced Level (A/L)[6]. The student's further education, and at times professional choices, are extremely reliant on performance at the said tests. Smart devices with facilities of delivering customized advice, flagging risks at an earlier point of time, and interventions at the due time are required for enhanced performance at school and level playing fields between urban and rural schools.

Of all of the Intelligent Education Support System's constituents, the Attendance Trend Analyzer has the critical task of analyzing and interpreting one of the stronger student success predictors—attendance. There are many studies which substantiate that those with stable attendance are academically more successful and habitual truancy has strong correlations with low achievement

and risk of dropping out. In Sri Lanka, wherein school and university policies will typically mandate at least an attendance rate of 80% for eligibility for exams, attendance comes to be considered more of an eligibility factor for exams and less of an informative resource.[6]

In practice, however, existing attendance monitoring practices are inadequate. A large proportion of the institutions use static percentages for calculating attendance without regard for trends, patterns, and context. For example, with an average of 75% attendance, a student will continue to have attendance lows on key weeks leading up to tests a scenario static thresholds will not measure.[7] Moreover, very few of the existing systems incorporate attendance with other behavioral or academic indicators like test scores, participation, or stress levels. As such, there emerges an immense knowledge gap as massive attendance data are being generated without being utilized effectively for the purpose of informing decisions.

Attendance Trend Analyzer fills this gap with time-series forecasting models (e.g., LSTM, ARIMA) and anomaly detection processes for managing attendance data. It spot trends, and detects abrupt irregularities, and predicts attendance risks for the future. Moreover, it relates attendance variations with academic performance as well as external events like test days or socio-economic interruptions.[8] Outputs are represented as graphical dashboards and reports, facilitating data-sourced decisions for administrators and teachers. Interpreting attendance records as leading indicators of concern, the Attendance Trend Analyzer offers an anticipatory tool for assisting students prior to the point of academic decline.[9] Its innovation has been expanding attendance from merely an indicator of responsiveness as far as being a leading and actionable academic indicator, and hence as part of the overall Intelligent Education Support System.[10]

1.1. Background & Literature Survey

Attendance has long been recognized as one of the most reliable indicators of student engagement and academic success. Students who regularly attend classes benefit from continuity in learning, direct teacher guidance, and peer collaboration, which collectively improve their performance in assessments. Conversely, chronic absenteeism is associated with weaker academic achievement, disengagement, and in severe cases, school dropout. Studies by the National Center for Education Statistics highlight that absenteeism contributes to long-term negative educational outcomes if not addressed early [11].

In Sri Lanka, the importance of attendance is further magnified due to the examination-driven education system. Performance in national-level assessments such as the Grade 5 Scholarship, G.C.E. Ordinary Level (O/L), and G.C.E. Advanced Level (A/L) examinations largely determines students' future educational and career opportunities. To maintain discipline and accountability, most schools and universities enforce attendance thresholds—commonly 80% minimum attendance—as eligibility criteria for examinations and course completion [12], [13]. While these policies formalize attendance as a compliance requirement, they do not provide meaningful insights into trends (e.g., gradual declines), critical risk periods (e.g., pre-exam absenteeism), or contextual anomalies (e.g., absenteeism linked to external disruptions like transport strikes).

Recent changes in student behavior further highlight the urgency for improved monitoring. Post-pandemic shifts toward flexible learning, financial pressures, and a growing reliance on digital resources have caused attendance volatility worldwide. Reports highlight a steady decline in lecture attendance and increased disengagement, which threaten teaching effectiveness and student outcomes [14]. At the same time, emerging technologies such as AI-powered facial recognition are increasingly used to automate attendance collection, producing more accurate datasets that can be analyzed for patterns and insights [115].

Despite these advancements, most Sri Lankan institutions continue to rely on simple percentage-based reporting. The challenge lies not in data availability but in the lack of advanced analytics to extract actionable insights from attendance logs. This gap forms the foundation for the Attendance Trend Analyzer, which transforms attendance records into a predictive and diagnostic tool for academic risk management.

The relationship between attendance and academic performance is well documented in global and regional research. Nguyen and Zhang (2023) confirm through econometric methods that regular attendance significantly improves student grades, with the strongest effects observed among lower-performing students [16]. Earlier studies by Marburger and Gunn (2016) also reported a direct correlation between attendance rates and examination scores [17]. These findings establish attendance as a robust predictor of academic outcomes, making it a suitable variable for predictive analytics.

In the field of **educational data mining and learning analytics**, attendance is widely incorporated into **Early Warning Systems (EWS)** that aim to identify at-risk students. Baker (2019) notes that

integrating attendance with academic and behavioral features can significantly improve dropout prediction accuracy [18]. Vasconcelos et al. (2023) extend this by demonstrating that data-driven models can anticipate dropout risks weeks or months before final assessments, enabling timely interventions [19]. However, in many systems, attendance is still treated as a **static feature** rather than a **dynamic time-series signal**, which limits its predictive power.

Recent advances in **time-series analysis** and **deep learning** have created opportunities to analyze attendance in new ways. Chen et al. (2024) survey time-series methods such as ARIMA, LSTM, and hybrid ARIMA-LSTM models, showing their effectiveness in forecasting educational behaviors [20]. Jain et al. (2024) propose an ensemble ARIMA-LSTM model that combines statistical rigor with neural network adaptability, achieving higher forecasting accuracy in education datasets [21]. Applying such models to attendance can uncover **hidden patterns** such as seasonal dips, pre-exam declines, and long-term disengagement trajectories.

Parallel to forecasting, **anomaly detection methods** are increasingly applied in educational contexts. Guo et al. (2022) document how density- and reconstruction-based anomaly detection algorithms can reveal unusual patterns in student behavior, often linked to academic risk or well-being concerns [22]. For attendance data, anomaly detection can identify sudden drops, irregular spikes, or group-level deviations that may signal emerging risks.

Closer to Sri Lanka, institutional studies confirm the policy importance of attendance but also expose operational gaps. For example, universities such as the Faculty of Medicine at Peradeniya enforce 80% attendance rules, while national-level reporting (*The Sunday Times, Ceylon Today*) reiterates similar requirements. Yet, few academic studies systematically analyze how **attendance trends over time** relate to academic achievement or how anomalies can guide interventions. This underscores the need for a tool like the Attendance Trend Analyzer, which integrates **forecasting, anomaly detection, and correlation analysis** into a user-friendly dashboard for teachers and administrators

1.2 Research Gap

There are many systems available to record and monitor student attendance. Some researchers have applied IoT devices, RFID, GPS, NFC, or even facial recognition to automate attendance tracking. Others have used data mining and simple analytics to explore attendance percentages or

dropout risks. But the prediction of attendance trends using time-series models, detection of anomalies, and correlation of attendance with academic calendars and performance factors has not been done. The proposed Attendance Trend Analyzer addresses this gap by introducing forecasting, anomaly detection, and contextual insights, turning attendance into a proactive early-warning tool for academic support.

a. Comparison of former research

Application References	Trend Analysis (Time-series)	Anomaly Detection	Predictive Modeling	Contextual Correlation	Visualization & Actionability
ERIC Report[23]	✓	✗	✓	✗	✓
IJAMTES, 2021[24]	✓	✓	✗	✗	✓
EDM, 2022[25]	✓	✗	✗	✗	✗
Frontiers, 2023[25]	✗	✗	✓	✗	✓
ResearchGate, 2023[26]	✓	✗	✓	✗	✗
ArXiv, 2023[27]	✗	✗	✗	✗	✓
ArXiv, 2023[28]	✗	✓	✗	✗	✗
Proposed: Attendance Trend Analyzer(ATA)	✓	✓	✓	✓	✓

Daily Attendance Time-Series Analysis (ERIC Report) [23]

The ERIC study on school attendance time-series analyzed daily attendance records from two high schools across an academic year, examining fluctuations and seasonality. While it demonstrated that attendance data can be modeled as a time series, the study was primarily descriptive and did not apply forecasting techniques or anomaly detection to predict risks. The proposed Attendance Trend Analyzer will extend this work by employing ARIMA and LSTM forecasting, anomaly detection, and correlation with academic events, making attendance a proactive early-warning tool rather than a static observational record.

Attendance Forecasting for Academic Counseling (IJAMTES, 2021) [24]

The IJAMTES paper explored predictive analytics for semester-level attendance, aiming to support counseling and intervention. Although it applied basic forecasting methods, the approach was limited in scope and lacked anomaly detection or contextual awareness. The proposed Attendance Trend Analyzer builds on this by integrating advanced forecasting (ARIMA/LSTM) with context-aware anomaly detection and academic calendar correlation, providing educators with more precise and actionable insights.

EMA + KLD Divergence Detection (EDM, 2022) [25]

The Educational Data Mining poster introduced a method using Exponential Moving Average (EMA) and Kullback–Leibler Divergence (KLD) to identify divergence points in attendance between high- and low-performing students. This approach highlights critical periods where performance differences emerge. However, it focuses only on divergence detection and does not deliver full predictive or contextual analytics. The proposed Attendance Trend Analyzer expands this by combining divergence detection with forecasting models, anomaly detection, and visual dashboards, enabling institutions to identify risks earlier and in richer context.

Dropout Prediction with Machine Learning (Frontiers, 2023) [26]

The Frontiers study [26] applied clustering and classification algorithms to predict undergraduate dropout risks using multimodal student data, including demographics and academic performance. While effective at identifying at-risk students, it did not specifically leverage attendance as a time-series variable nor integrate contextual factors. The proposed Attendance Trend Analyzer

addresses this gap by treating attendance as a dynamic, sequential signal, applying predictive models, and linking attendance dips with exam schedules and external disruptions to provide more targeted academic risk alerts.

Flipped-Classroom Attendance Analytics (ResearchGate, 2023) [27]

A research effort in flipped-classroom environments used LMS activity and participation logs to monitor engagement and predict risks. While valuable in digital learning contexts, it was limited to a specific pedagogical model and did not directly analyze in-class attendance trends or anomalies. The proposed Attendance Trend Analyzer generalizes this idea, combining traditional attendance records with LMS data, applying time-series forecasting and anomaly detection, and contextualizing attendance with exams and holidays for broader applicability in Sri Lankan schools and universities.

IoT–RFID Attendance System Review (ArXiv, 2023) [28]

The IoT–RFID review surveyed systems that use RFID and IoT devices for automated attendance tracking. These systems improve accuracy and efficiency in logging attendance but remain limited to verification and automation, without predictive modeling or academic integration. The proposed Attendance Trend Analyzer builds on this foundation by adding forecasting capabilities, anomaly analytics, and contextual correlation, transforming attendance from a verification tool into an academic decision-support system.

Attendance Data Credibility for Mining (ArXiv, 2023) [29]

The Attendance Data Credibility study [12] proposed methods to assess and improve the reliability of attendance logs by introducing a Student Attendance Credibility (SAC) score. While this improves data quality for educational data mining, it does not address forecasting, anomaly detection, or contextual insights. The proposed Attendance Trend Analyzer complements such credibility frameworks by applying predictive modeling, anomaly detection, and contextual correlation, ensuring that clean attendance data translates into actionable academic insights for teachers and administrators.

1.3. Research Problem

Despite the recognized importance of students' attendance in shaping academic performance [30], most Sri Lankan schools still rely on traditional manual attendance tracking methods that only record presence or absence without providing deeper analysis[31]. These approaches fail to identify patterns. Recurring trends, or signals of irregularity. As a result, teachers and administrators often intervene reactively after students already show signs of academic decline rather than proactive detecting risks in advance [32].

Furthermore, the lack of predictive modeling and anomaly detection in existing systems prevents schools from forecasting future attendance issues or identifying students with usual absenteeism behavior [33]. Without visual and actionable insights, administrators face difficulties in linking attendance data with academic outcomes, making it challenging to support evidence-based decision-making for performance improvement [34].

Therefore, there is a pressing need to develop a data-driven Attendance Trend Analyzer that can

- Identify patterns and trends in attendance behavior,
- Apply predictive analytics to forecast future absenteeism,
- Detect anomalies and irregularities in attendance, and
- Provide visualizations and reports that enable teachers and school leaders to take proactive action to improve student outcomes in the Sri Lankan education context. [35]

2. OBJECTIVES

2.1. Main Objectives

To develop an AI-powered Intelligent Education Support System that enhances personalized learning, predicts academic risks, monitors attendance trends, and supports student well-being through data-driven insights, thereby improving overall academic performance and enabling timely interventions for Sri Lankan students.

2.2. Specific Objectives

1, Data Collection & Preprocessing

- To gather and preprocess student attendance from selected schools in Sri Lanka for accurate analysis.

2. Trend Analysis

- To analyze historical attendance data, identify recurring patterns, seasonal trends, and long-term behaviors.

3. Predictive Modeling

- To apply time-series forecasting models (e.g, ARIMA, LSTM)for predicting future attendance trends)

4. Anomaly Detection

- To detect irregular attendance behaviors and highlight students with unusual absenteeism.

5. Visualization and Reporting

- To design interactive dashboards and visualization tools that help teachers and administrators interpret attendance data easily.

6. Correlation with Academic Outcomes

- To investigate the relationship between attendance trends and academic performance to identify at risk students

7. Localized Implementation

- To develop a solution tailored to the Sri Lankan educational context for practical adoption and decision-making

3.METHODOLOGY

The methodology of the AI-Powered Intelligent Education Support System is designed as a multi-component framework, integrating data collection, preprocessing, machine learning, and

visualization techniques. The system is modular, with each component focusing on a unique educational challenge while sharing a unified data pipeline and decision-support architecture.

3.1. System Architecture

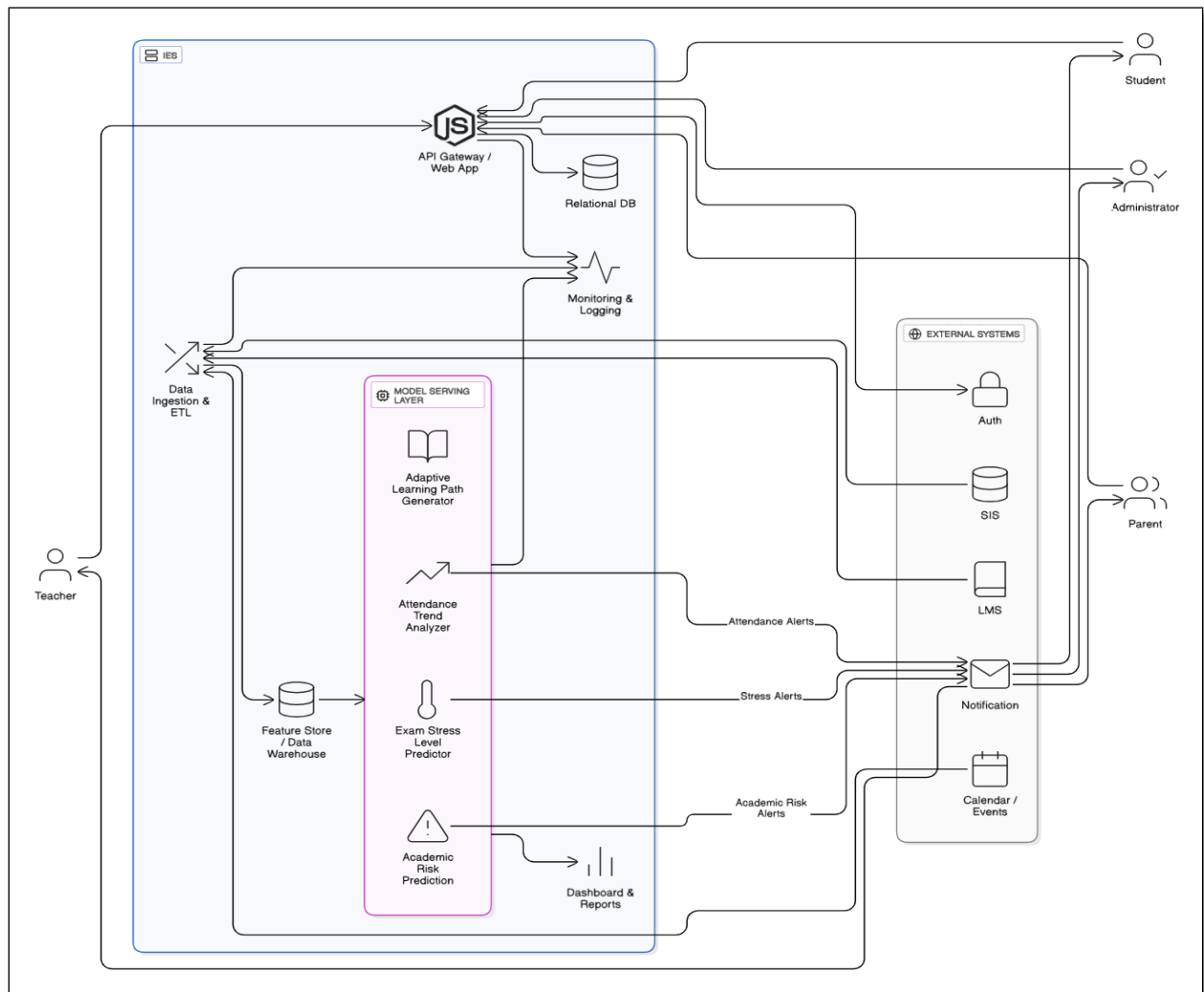


Figure 3.1: Overall system diagram

Figure 3.1 illustrates the overall high-level diagram of the proposed component, which analyzes student attendance to forecast risk and trigger timely interventions.

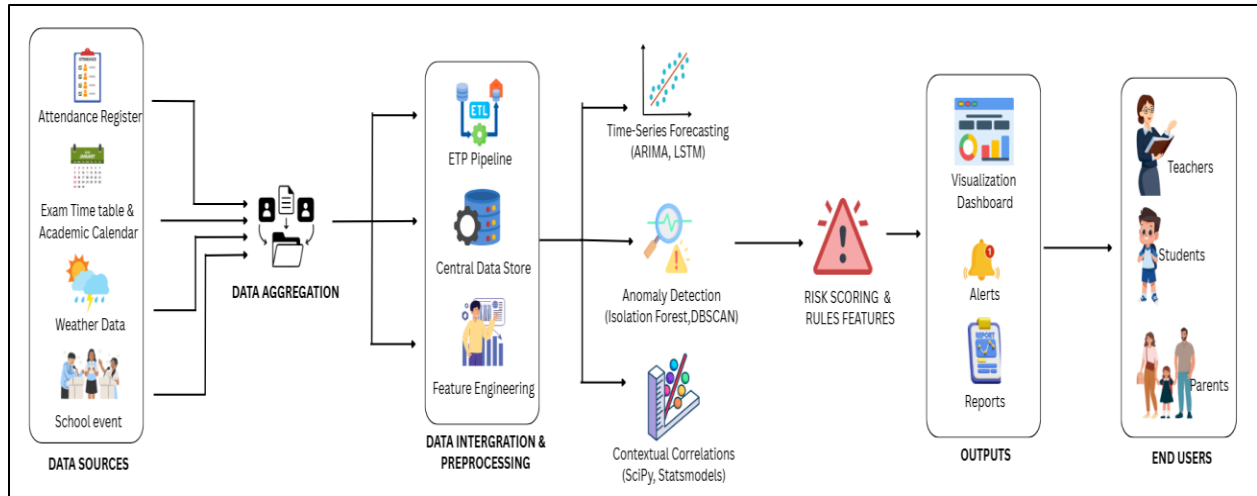


Figure 3.2: Attendance Trend Analyzer

Initially, the stakeholder (teacher/administrator) signs in to the cross-platform dashboard (web/mobile). The stakeholder can either upload attendance CSVs or connect the Learning Management System (LMS)/Student Information System (SIS) to stream records automatically. The ingested data are serialized (JSON/CSV) and sent to a Node/Express API, which stores them in the Student Records Database and forwards cleaned batches to a Python/Flask ML service. Inside the ML service, the data are preprocessed (time indexing, missing-value handling, normalization), then passed to time-series models to forecast near-term attendance and to anomaly detectors to catch sudden irregularities. The service also performs contextual correlation with exam timetables, holidays, and recent academic performance. The resulting risk scores, explanations, and trends are returned to the backend and rendered on the dashboard as heatmaps, charts, and reports. Alerts are pushed to staff when risk thresholds are exceeded, along with recommended interventions (e.g., counseling, parental outreach, remedial sessions).

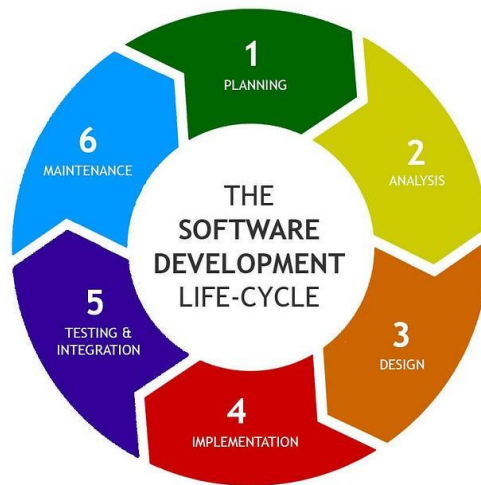
There are three subcomponents of the proposed component, as shown in Figure 3.2. First, the system performs trend forecasting (ARIMA/LSTM) to predict attendance dips before critical periods. Afterward, the system applies anomaly detection to identify sudden drops or unusual cohort patterns that static percentages miss. Finally, the system conducts contextual correlation and risk scoring, linking trends and anomalies with assessment schedules and external events to determine academic impact. After identifying risk, the module issues severity-based alerts and provides actionable recommendations for timely intervention.

Table 3.1: Technologies, techniques, architecture, and algorithms used

Technologies	Python, MERN Stack, Dash, TailwindCSS, Power BI, GitHub
Techniques	Machine Learning, Data Processing, Correlation Analysis, Data Visualization
Algorithms	ARIMA, LSTM, DBSCAN, Pearson, Threshold-based Alerting
Architectures	RNN, LSTM, Bi-LSTM, GRU

3.1.1. Software Solutions

The Software Development Life Cycle (SDLC) is a systematic approach that guides development teams through the planning, designing, constructing, testing, and delivering of quality software. It



ensures that every phase of development—from requirement analysis, planning, design, implementation, testing, deployment, to maintenance—is carried out in a structured way to minimize risks and meet customer expectations. Traditional models such as the Waterfall model follow a linear sequence in which one phase must be completed before moving to the next, but modern methodologies such as Agile are adaptive and allow teams to react to changing requirements. By following SDLC, organizations can make sure that they are delivering software that is reliable, within budget, and aligned with users' needs.

Figure 3.3: SDLC Life cycle

1. Data Gathering

Attendance records, academic schedules, and examination data were collected from partner institutions. Where LMS systems existed, datasets were exported digitally. For institutions still relying on manual registers, sample data were collected manually and digitized.

- Stakeholder Feedback

Surveys and interviews were conducted with teachers, administrators, and students to understand practical issues in attendance tracking, limitations of current systems, and requirements for an effective early-warning tool.

- Problem Definition

Stakeholders highlighted that existing systems only report attendance percentages but cannot forecast trends, detect anomalies, or provide actionable insights. This clarified the scope for predictive analytics and visualization features.

2. Feasibility Study (Planning)

- Economic Feasibility

The solution is developed using open-source technologies such as Python, Flask, TensorFlow, PostgreSQL, and React, minimizing software costs. Deployment can be cloud-based or local, making it affordable for schools with limited budgets.

- Scheduled Feasibility

The project timeline was divided into sprints of 2–3 weeks. Each sprint targeted deliverables like data preprocessing, ARIMA/LSTM forecasting, anomaly detection, visualization dashboards, and integration with the Intelligent Education Support System.

- Technical Feasibility

Development requires expertise in machine learning (time-series forecasting, anomaly detection), data pipelines, back-end APIs, and interactive dashboards. The team possesses the required skills and resources to implement the system.

3. Design

- System Design

The architecture consists of four layers: Data Collection, Data Processing & ML, Visualization & Dashboard, and User Interaction. Each layer communicates through APIs and a centralized student records database.

Subcomponents include:

- Trend Forecasting Module (ARIMA/LSTM)
- Anomaly Detection Module
- Correlation Analysis with Exams & Events
- Visualization Dashboards (Heatmaps, Charts, Alerts)

Use Case Diagram

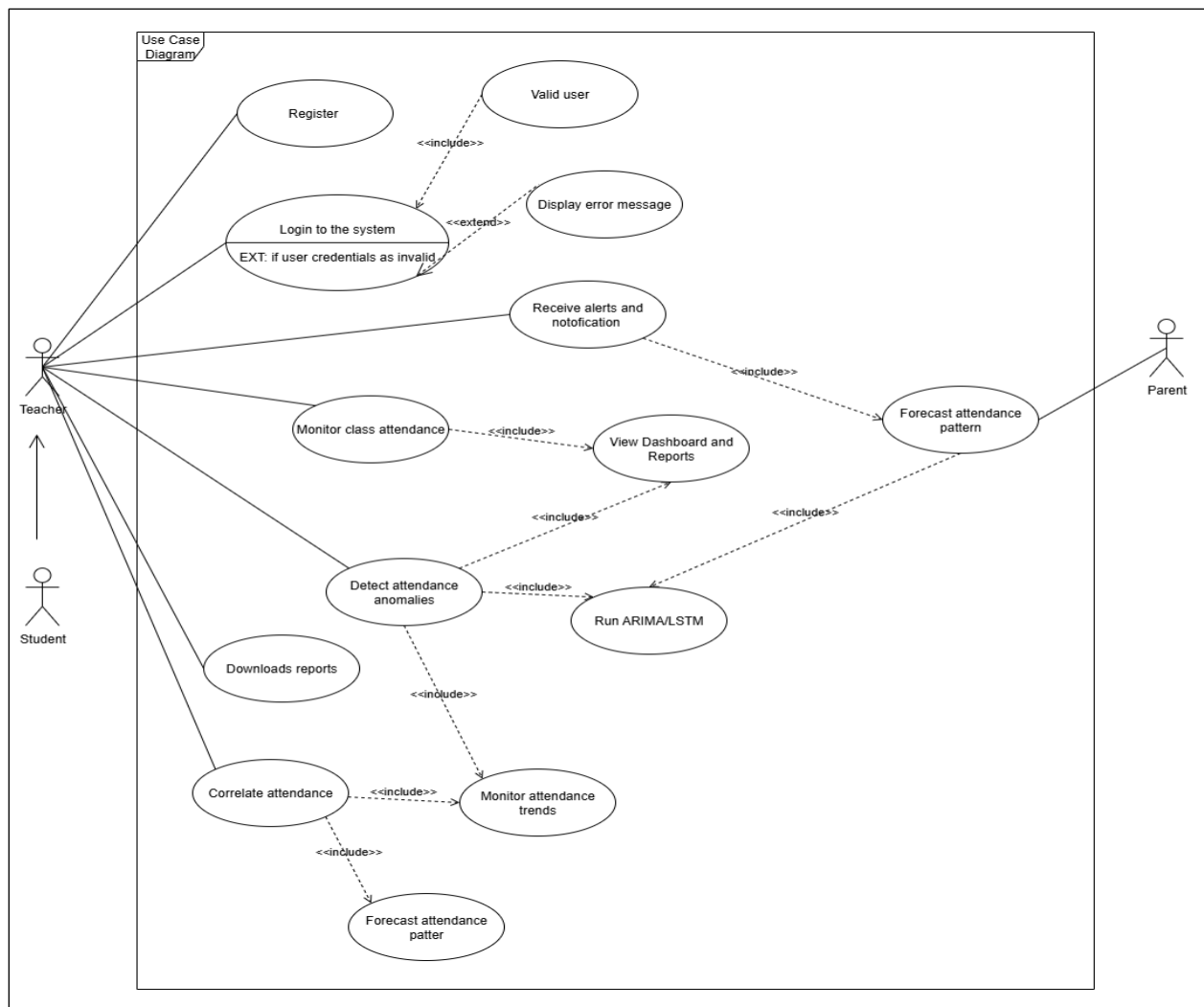


Figure 3.4 : Use Case diagram

Sequence Diagram

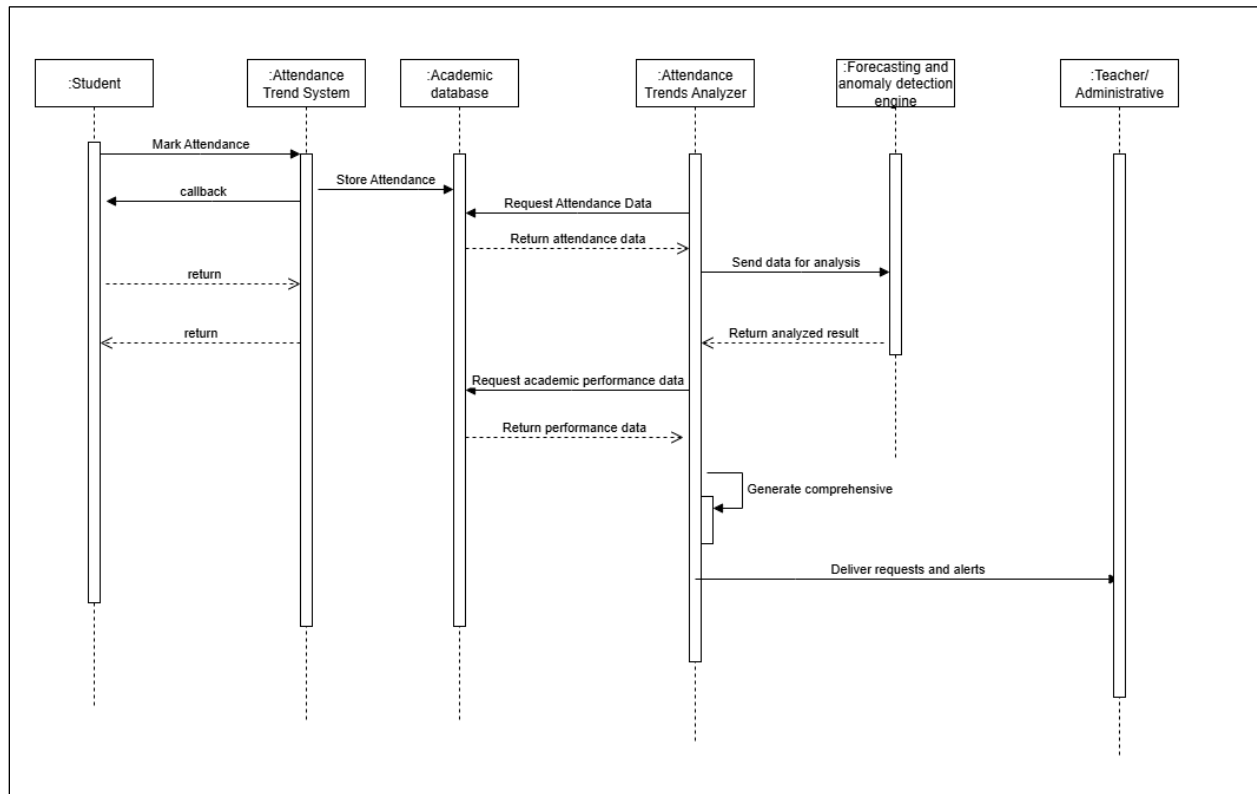


Figure 3.5 : Sequence Diagram

- Database Design

A relational schema was designed to store attendance logs, exam results, and risk scores. Indexing and caching ensure fast retrieval for dashboards.

- Interface Design

A responsive web interface for teachers/administrators provides real-time alerts, reports, and actionable recommendations.

4. Implementation

- Development followed Scrum sprints, with each sprint focusing on a functional module.
- Python was used for ML algorithms (ARIMA, LSTM, anomaly detection) with Pandas and NumPy for preprocessing.
- Flask microservices exposed ML predictions as APIs to the Node.js backend.

- React.js was used to build dashboards, charts, and heatmaps for visualization.
- GitHub was used for version control, while Jira/Trello tracked sprint progress and backlog items.

5. Testing

- Unit Testing: Individual modules such as forecasting models, anomaly detection, and correlation analysis were tested using Python's unittest and PyTest frameworks.
- Integration Testing: Verified smooth communication between data pipelines, ML microservices, and dashboards.
- System Testing: End-to-end tests were run to validate accuracy of attendance forecasting and correctness of anomaly alerts.
- User Acceptance Testing (UAT): Teachers and administrators evaluated the dashboard prototypes and provided feedback for improvements.

5. Deployment

- The system was deployed on a cloud platform (e.g., AWS/Heroku) for scalability.
- Docker containers were used to package ML services and back-end APIs, ensuring portability.
- Continuous Integration/Continuous Deployment (CI/CD) pipelines automated builds, tests, and deployments.
- Secure login (JWT/OAuth2) ensured data privacy for institutions.

6. Maintenance

- Regular updates will be provided to refine ML models with new attendance data, improving accuracy.
- Logs and performance metrics will be monitored to detect bottlenecks or failures.
- Feedback from teachers and administrators will be incorporated into sprint cycles for continuous improvement.
- Periodic retraining of ARIMA and LSTM models ensures they remain effective as new data trends emerge.

3.1.2. Commercialization and Business plan

The proposed Attendance Trend Analyzer would be sold as an education technology solution to aid schools, colleges, and parents in monitoring and improving the performance of students. The system would be available as a web application as well as a mobile application to enable easy use by teachers, administrators, parents, and students. It will come in two flavors: a basic one with features such as attendance collection, trend forecasting, and abnormality identification, and a professional one with additional features such as parent alerts, contextual correlation with test timings, and in-depth reports.

Market target includes schools, colleges, private tuition centers, ministries of education, and research institutes. Parents and students will also gain through reminders and information that promote accountability and early intervention. The system is designed to be simple to use, without technical skill required, and thus can reach a wide audience.

The business model will be subscription-based, where institutions pay a subscription fee depending on the number of students annually or monthly. Premium subscriptions will earn revenue through higher-end features such as automated alerts and integration with LMS. The cost will mainly be cloud hosting, storage, and sending notifications, which will be minimal thanks to the utilization of open-source technologies.

To popularize adoption, the system will be marketed through pilot initiatives in schools, demo runs, web promotions, and partnerships with education authorities. With heightened awareness, larger institutional and governmental alliances can push the system to broader audiences. Economically, the project is sustainable because even a small segment of school subscriptions will pay for themselves and provide funds for expansion.

The system can scale up to national and even international levels in the future, with advantages being reaped by education ministries and large institutions through a data-driven attendance monitoring system, boosting student achievements.

3.1.3. Future scope

In the future, Attendance Trend Analyzer can be enhanced with more advanced features such as AI-enabled intervention planning, where the system not only alerts risk but also recommends individualized solutions such as remedial classes or counseling based on student profiles. Multimodal learning analytics (LMS activity, level of engagement, and behavioral cues) integration will allow for holistic comprehension of student performance. In addition, the integration with Explainable AI (XAI) will make predictions understandable to teachers and parents, promoting adoption and confidence. At a broader level, the system can enable cross-institutional benchmarking and national policy-making by providing aggregated insights into absenteeism trends, enabling evidence-based measures to improve education quality.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

They outline the system's key features. They are necessary for the purpose of making the system effectively capture, process, review, and display attendance information.

- Import student attendance records from schools via manual uploading (CSV/Excel).
- Handling and cleaning data with missing values, data duplication, and inconsistencies in formatting.
- Analyze attendance records for patterns between and between students, courses, subjects, and periods of learning.
- Utilize forecasting models like ARIMA and LSTM for predicting attendance trends for future days.
- Defects, irregularities, and aberrant patterns of absentee
- Link attendance records with academics, socioeconomic status, health status, and parental participation. And school activities.
- See attendance trends with interactive dashboards, heatmaps, and charts.
- Flag college student alerts automatically below required attendance levels.
- Implement role-based access for administrators, teachers, and parents.

- Export attendance records as PDF, Excel, or CSV files
- Maintain historical attendance records for longitude analysis.

4.2. Non-functional Requirements

- Process large data sets quickly with many students, courses, and semesters.
- Scaling up to many schools, grades, and increasing data sets without performance loss.
- Offer a simple and intuitive user interface which will allow those with barely any technical experience.
- Safeguard confidential information, e.g., attendance, grades, socio-economic data, and health information.
- Protect data privacy and manage access for personal student data.
- Offer reliable and consistent performance with low down time.
- Offer access on both browsers and cell phones.
- Interface with existing school administration system and any other external sources of data.
- Ensure data integrity with accuracy, consistency, and completeness of records.

4.3. System Requirements

The proposed Attendance Trend Analyzer requires the following software technologies and tools for deployment:

1. VS Code – As the primary Integrated Development Environment (IDE) used to develop and edit Python and JavaScript code.
2. Flask Server – For serving and running the machine learning models for predicting attendance, detecting anomalies, and contextual correlations.
3. React / React Native – To create an interactive cross-platform web and mobile application for teachers, administrators, and students.
4. Python – To be used for backend development and running machine learning algorithms (ARIMA, LSTM, anomaly detection).
5. PostgreSQL / MySQL – Database management system to store attendance data, academic performance, and risk alerts.

6. Chart.js / Plotly – For data visualization, heatmaps, and generating reports in the dashboard.
7. Docker (optional) – For containerizing the ML models and backend services.

4.4. User Requirements

The Attendance Trend Analyzer will be utilized by different stakeholders within the education system. Each type of user will have specific requirements and functionality:

Teachers / Lecturers

- Upload or sync student attendance data through the system.
- View dashboards, heatmaps, and attendance predictions for their classes.
- Be notified of students with poor or declining attendance.
- Export attendance reports for academic or administrative purposes.

Administrators / School Management

- Monitor institution-level attendance trends across classes and departments.
- Correlate attendance with exam timetables and academic performance.
- Establish system thresholds and policy for tracking attendance.
- Generate detailed reports for policy-making and academic review.

Students

- See their own attendance record and forecasts.
- Receive reminders and encouragement to improve attendance consistency.
- View how their attendance is impacting academic performance.

Parents / Guardians

- Receive notifications and alerts when their child's attendance varies significantly.
- View attendance summaries through a mobile-friendly portal.
- Interact with teachers and school administrators when their child has been flagged as at risk.

4.5. Wireframes

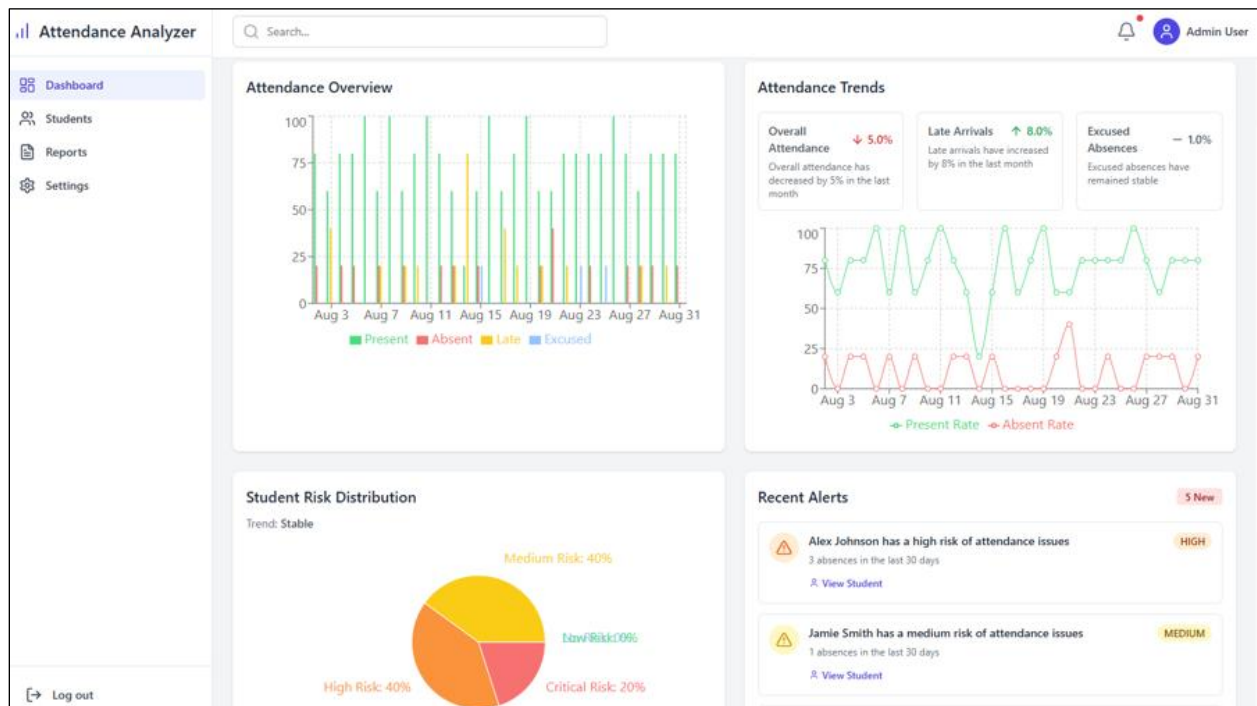


Figure 4.1 : Attendance Trends Analyzer Dashboard

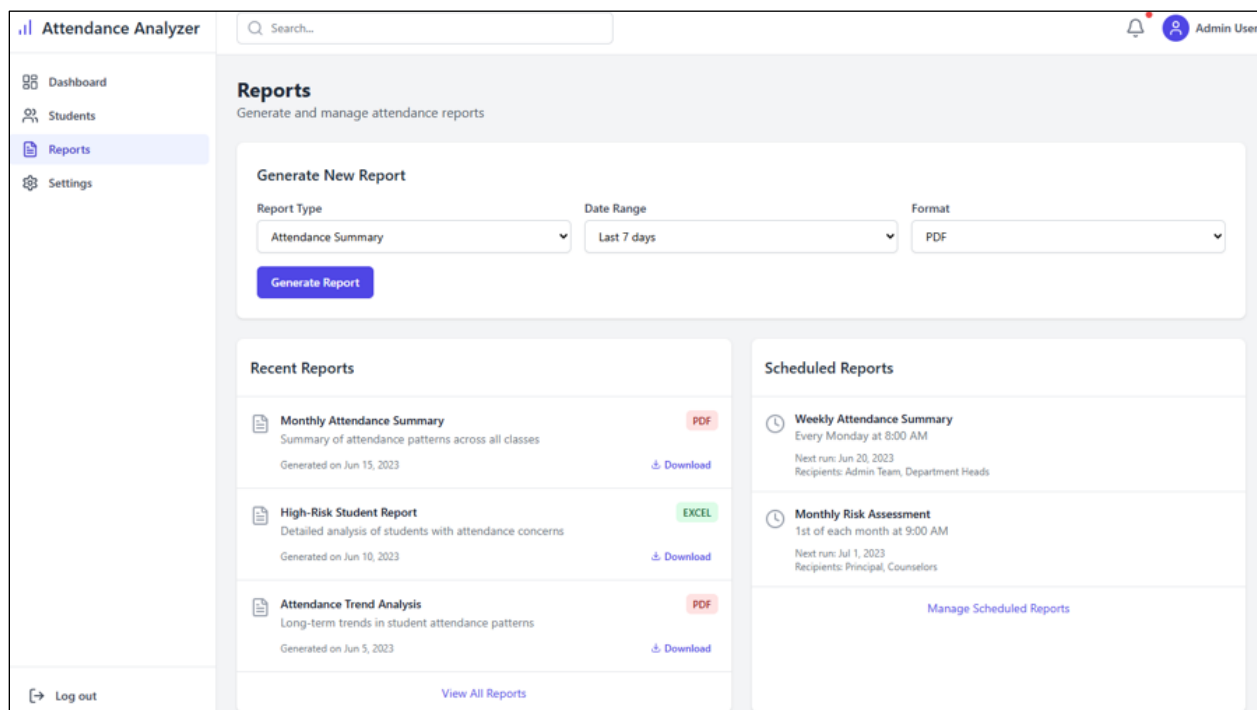


Figure 4.2 : Attendance Trend Analyzer Report Page

Attendance Analyzer

Search...

Admin User

Dashboard

Students

Reports

Settings

Configure when alerts should be triggered based on risk levels

Critical Risk Threshold

50

High Risk Threshold

30

Notification Channels

Select which channels to use for different alert types

☒ Email

Send email notifications for alerts

☒ In-App

Show alerts in the dashboard

☐ SMS

Send text message alerts for critical risks

Data Sources

Attendance Management System

Last sync: 2 hours ago

Connected

Learning Management System

Last sync: 1 day ago

Connected

Academic Database

Last sync: 3 hours ago

Connected

Calendar/Exam Service

Not configured

Disconnected

Configure Integrations

Visualization Settings

Default Chart Type

Bar Chart

Color Scheme

Default

☒ Show Legends

Display legends on all charts

Apply Changes

Log out

Figure 4.3 : Attendance Trend Analyzer Setting Page

5. GRAND CHART

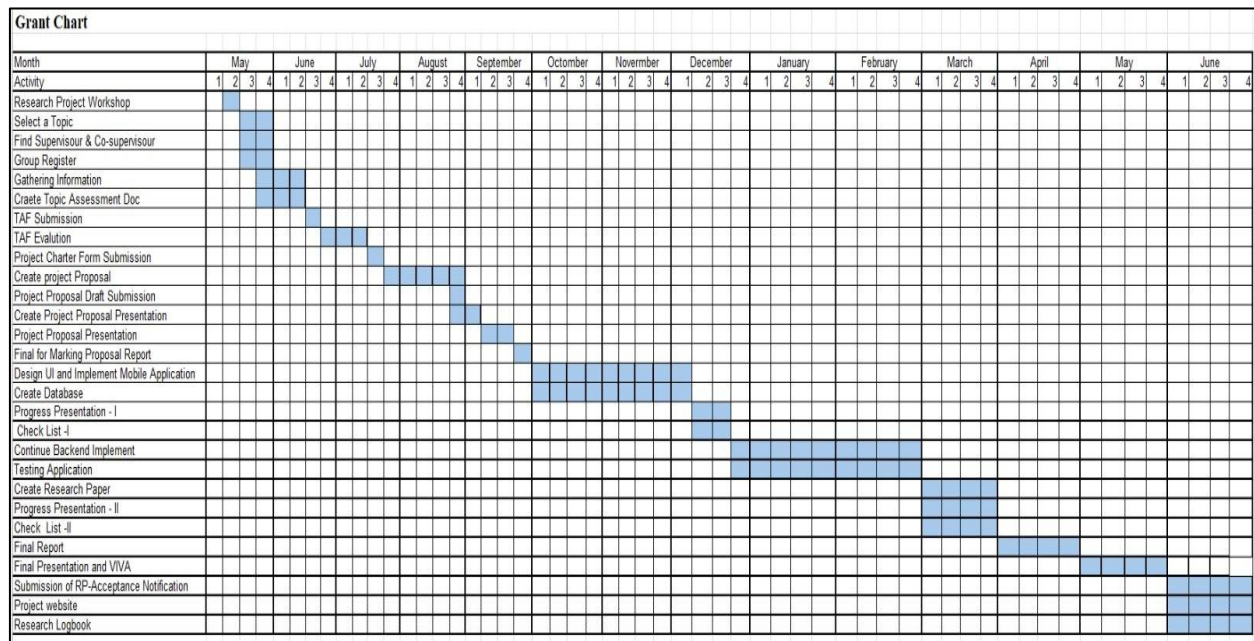


Figure 5.1: Gantt chart

5.1. Work Breakdown Structure(WBS)

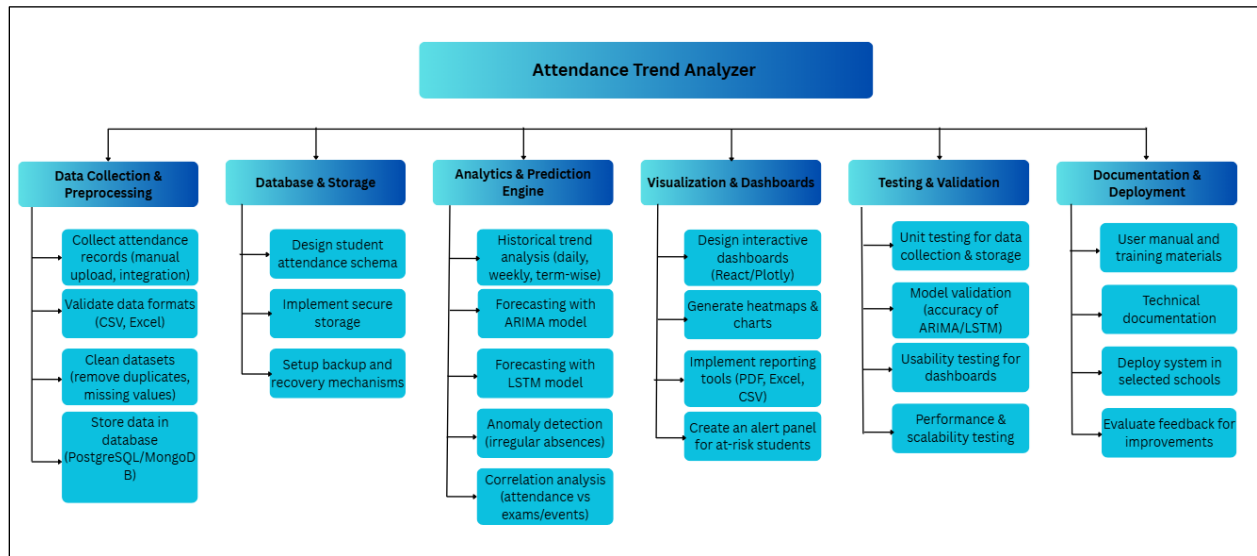


Figure 5.2: Work Breakdown Chart of Attendance Trend Analyzer

6. BUDGET AND JUSTIFICATION

Below table 6.2 shows the overall budget for the overall proposed system.

Table 6.1: Expenses for the proposed system

Expenses	
Requirement	Cost (Rs)
Travelling cost for field visit and data collection	30,000.00
Deployment cost	15,799.00 per month
Hosting cost	10,000.00
Total Expenses	55,799.00

REFERENCES

- [1] J. Tilak, "Education and development," in *International Handbook of Educational Research in the Asia-Pacific Region*, J. Keeves & R. Watanabe, Eds. Dordrecht: Springer, 2003, pp. 93-110. doi: 10.1007/978-94-017-3368-7_7.. Accessed date: July 31, 2025
- [2] A. Tilli et al., "The changes in the effects of COVID-19 on education: A meta-narrative review," *Tech., Knowl. and Learn*, pp. 1-29, 2023. doi: 10.1007/s10758-023-09684-4. Accessed: Aug. 20, 2025. [Online].
- [3] M. Janssen et al., "Will the digital divide disappear? An analysis of the changing digital divide in Europe, 2010-2019," *Gov. Inf. Quarterly*, vol. 39, no. 4, Oct. 2022, Art. no. 101756. doi: 10.1016/j.giq.2022.101756.
- [4] B. du Boulay, "Artificial intelligence as an effective classroom assistant," *IEEE Intell. Syst.*, vol. 31, no. 6, pp. 76-81, Nov. 2016. doi: 10.1109/MIS.2016.93.
- [5] C. Conati et al., "Artificial intelligence for education," in *Handbook of Artificial Intelligence*, 2nd ed. Boca Raton, FL, USA: Chapman and Hall/CRC, 2022, ch. 25, pp. 787-822.
- [6] S. J. Hettige, "Education, equity and social cohesion: A Sri Lankan case study," in *Education and Social Cohesion in a Post-Conflict and Divided Nation*, S. P. Heyneman, Ed. Rotterdam: SensePublishers, 2017, pp. 87-106. doi: 10.1007/978-94-6300-935-5_6.
- [7] P. Athukorala and D. Senewiratne, "Digital divide and regional inequality in Sri Lanka: A spatial analysis," *Asia & the Pacific Policy Studies*, vol. 9, no. 2, pp. 168-186, May 2022. doi: 10.1002/app5.346.
- [8] C. M. Kearney, "Student absenteeism: A review and meta-analysis of its impact on academic achievement," *Educ. Psychol. Rev.*, vol. 34, no. 4, pp. 1-30, Dec. 2022. doi: 10.1007/s10648-022-09717-x. Accessed: Aug. 15, 2023. [Online]. Available: <https://link.springer.com/article/10.1007/s10648-022-09717-x>
- [9] Ministry of Education, Sri Lanka. *Circular No. 2022/08 - School Attendance and Exam Eligibility*. [Online]. Available: <https://www.moe.gov.lk/circulars/> (Accessed: Aug. 25, 2023).

- [10] A. P. Silva and M. M. T. Rodrigo, "Leveraging educational data mining for analyzing student attendance and its correlation with performance," in *Proc. 9th Int. Conf. Learn. Anal. & Knowl. (LAK19)*, Tempe, AZ, USA, 2019, pp. 391-395. doi: 10.1145/3303772.3303825.
- [11]. Ministry of Education Sri Lanka. (2019). School Attendance and Performance Report. Colombo: Government of Sri Lanka.
- [12]. Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and Practice* (2nd ed.). OTexts.
- [13]. Kearney, C. A. (2008). School absenteeism and school refusal behavior in youth: A contemporary review. *Clinical Psychology Review*, 28(3), 451–471.
- [14]. World Bank. (2020). *Transforming Education in Sri Lanka: From Vision to Action*. Washington, DC: World Bank.
- [15]. R. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 10, no. 3, pp. 1–27, 2020.
- [16]. V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.
- [17]. M. B. N. Nowzath and M. I. M. Riyath, "Evaluating the relationship between student attendance and academic performance," 8th South Eastern University International Arts Research Symposium, Dec. 2019.
- [18]. M. H. A. Munas, N. M. H. Mohamed, and S. M. C. Sithy, "The Impact of Students' Attendance on Effective Learning: An Evaluative Study," *European Journal of Humanities and Social Sciences*, 2019.
- [19]. S. Few, *Now you see it: Simple visualization techniques for quantitative analysis*, Analytics Press, 2009.
- [20]. J. Finn and K. Zimmer, "Student engagement: What is it? Why does it matter?" *Review of Educational Research*, vol. 71, no. 2, pp. 123–150, 2001.

- [21]. T. Jayawardena, “Digital transformation in Sri Lankan schools: Challenges in data-driven education,” *Journal of ICT in Education*, vol. 5, no. 1, pp. 25–39, 2020.
- [22]. K. Perera and M. Fernando, “Predictive analytics in Sri Lankan education: Opportunities and challenges,” *Sri Lanka Journal of Advanced Research*, vol. 7, no. 1, pp. 45–56, 2019.
- [23] ERIC Report, “Analyzing Daily Attendance Time-Series in High Schools,” 2014.
- [24] IJAMTES, “Attendance Forecasting for Academic Counseling,” 2021.
- [25] Educational Data Mining (EDM), “EMA + KLD for Longitudinal Attendance Divergence,” 2022.
- [26] Frontiers in Education, “Predicting Undergraduate Dropout with ML,” 2023.
- [27] ResearchGate Preprint, “Tracking Attendance in Flipped Classrooms Using Analytics,” 2023.
- [28] ArXiv Preprint, “IoT-RFID Attendance System Review,” 2023.
- [29] ArXiv Preprint, “Student Attendance Credibility Metric for Data Mining,” 2020.
- [30] V. Kassarnig, A. Bjerre-Nielsen, E. Mones, S. Lehmann, and D. Dreyer Lassen, “Class attendance, peer similarity, and academic performance in a large field study,” *PLOS ONE*, vol. 12, no. 11, pp. 1–23, Nov. 2017.
- [31] H. Perera and M. de Silva, “Manual attendance systems in Sri Lankan schools: Challenges and limitations,” *International Journal of Education Development in Sri Lanka*, vol. 6, no. 2, pp. 45–56, 2021.
- [32] R. S. Baker, “Educational data mining: An advance for intelligent systems in education,” *IEEE Intelligent Systems*, vol. 29, no. 3, pp. 78–82, 2014.
- [33] S. Jain, R. Kumar, and A. Saxena, “A novel ensemble ARIMA–LSTM approach for forecasting educational time-series data,” *Future Generation Computer Systems*, vol. 142, pp. 182–194, Jan. 2024.
- [34] T. Guo, X. Liu, and J. Li, “Educational anomaly analytics: Features, methods, and directions,” *Frontiers in Big Data*, vol. 5, Art. 811840, pp. 1–10, 2022.
- [35] M. Jarbou, H. Zedan, S. Aljazzar, and M. Ababneh, “Deep learning-based school attendance prediction for autistic students,” *Scientific Reports*, vol. 12, Art. 21380, pp. 1–12, Dec. 2022.

APPENDICES

Plagiarism Report

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